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Assignment 6 : Use the diabetes data set from UCI and Pima Indians Diabetes data set for performing the following: a. Univariate analysis: Frequency, Mean, Median, Mode, Variance, Standard Deviation, Skewness and Kurtosis b. Bivariate analysis: Linear and logistic regression modeling c. Multiple Regression analysis d. Also compare the results of the above analysis for the two data sets. Dataset link: https://www.kaggle.com/datasets/uciml/pima-indiansdiabetes-database

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

from sklearn.linear_model import LogisticRegression, LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, mean_squared_error, r2_score, confusion_matrix
```

```
pima = pd.read_csv("diabetes.csv")

print("Pima Dataset Shape:", pima.shape)
print(pima.head())
```

```
Pima Dataset Shape: (768, 9)
   Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin   BMI  \
0            6      148             72             35        0  33.6
1            1       85             66             29        0  26.6
2            8      183             64              0        0  23.3
3            1       89             66             23       94  28.1
4            0      137             40             35      168  43.1

   DiabetesPedigreeFunction  Age  Outcome
0                     0.627   50        1
1                     0.351   31        0
2                     0.672   32        1
3                     0.167   21        0
4                     2.288   33        1
```

```
from sklearn.datasets import load_diabetes
diabetes = load_diabetes()
sklearn_df = pd.DataFrame(diabetes.data, columns=diabetes.feature_names)
sklearn_df['target'] = diabetes.target

print("Sklearn diabetes Shape :", sklearn_df.shape)
print(sklearn_df.head())
```

```
Sklearn diabetes Shape : (442, 11)
        age       sex       bmi        bp        s1        s2        s3  \
0  0.038076  0.050680  0.061696  0.021872 -0.044223 -0.034821 -0.043401
1 -0.001882 -0.044642 -0.051474 -0.026328 -0.008449 -0.019163  0.074412
2  0.085299  0.050680  0.044451 -0.005670 -0.045599 -0.034194 -0.032356
3 -0.089063 -0.044642 -0.011595 -0.036656  0.012191  0.024991 -0.036038
4  0.005383 -0.044642 -0.036385  0.021872  0.003935  0.015596  0.008142

         s4        s5        s6  target
0 -0.002592  0.019907 -0.017646   151.0
1 -0.039493 -0.068332 -0.092204    75.0
2 -0.002592  0.002861 -0.025930   141.0
3  0.034309  0.022688 -0.009362   206.0
4 -0.002592 -0.031988 -0.046641   135.0
```

Univariate analysis: Frequency,Mean, Median, Mode, Variance, Standard Deviation, Skewness and Kurtosis

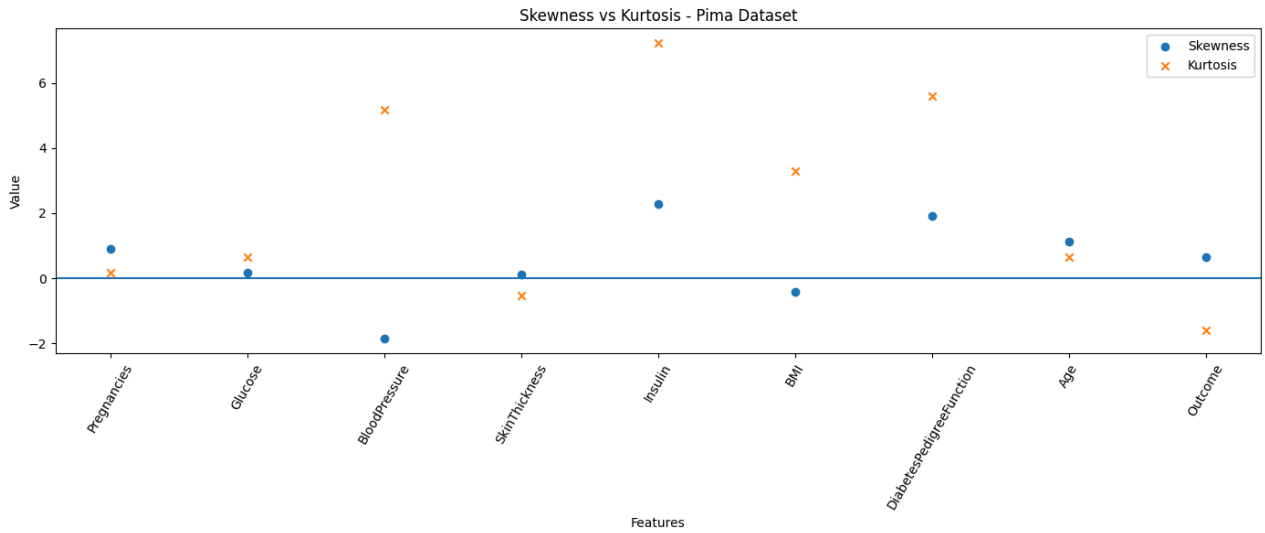
```
def univariate_analysis(df):
  result=pd.DataFrame({
      "Mean":df.mean(),
      "Median":df.median(),
      "Mode":df.mode().iloc[0],
      "Variance":df.var(),
      "Standard Deviation":df.std(),
      "Skewness":df.skew(),
      "Kurtosis":df.kurtosis()
  })
  return result
```

```
print("univariate analysis of pima dataset")
print(univariate_analysis(pima))

print("univariate analysis of sklearn diabetes dataset")
print(univariate_analysis(sklearn_df))
```

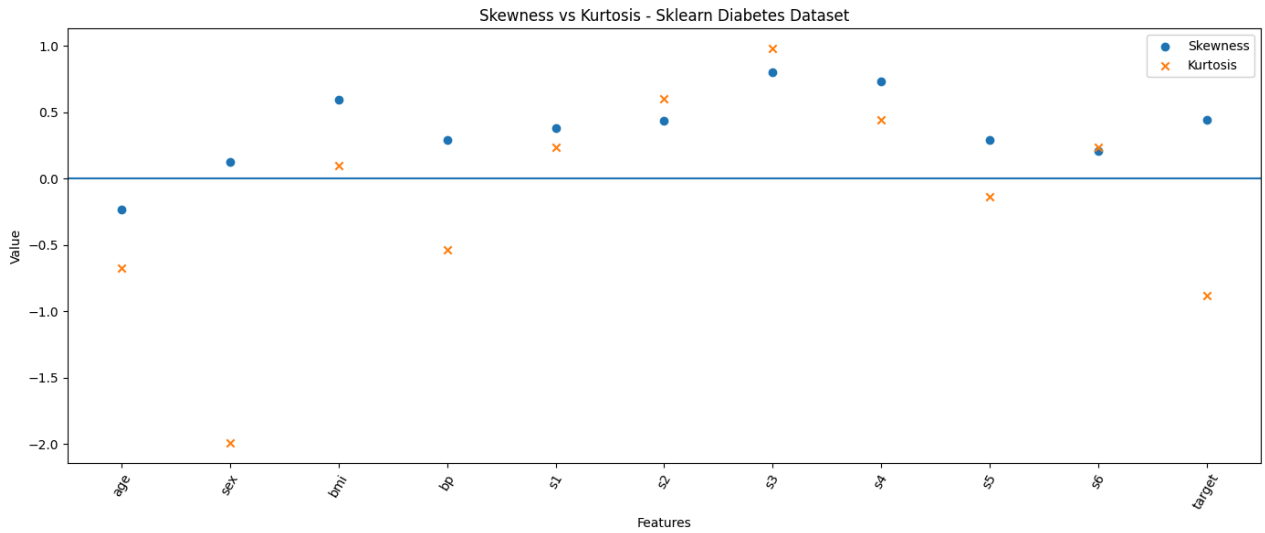
```
univariate analysis of pima dataset
                                Mean    Median    Mode      Variance  \
Pregnancies                 3.845052    3.0000   1.000     11.354056
Glucose                   120.894531  117.0000  99.000   1022.248314
BloodPressure              69.105469   72.0000  70.000    374.647271
SkinThickness              20.536458   23.0000   0.000    254.473245
Insulin                    79.799479   30.5000   0.000  13281.180078
BMI                        31.992578   32.0000  32.000     62.159984
DiabetesPedigreeFunction    0.471876    0.3725   0.254      0.109779
Age                        33.240885   29.0000  22.000    138.303046
Outcome                     0.348958    0.0000   0.000      0.227483

                          Standard Deviation  Skewness  Kurtosis
Pregnancies                        3.369578  0.901674  0.159220
Glucose                           31.972618  0.173754  0.640780
BloodPressure                     19.355807 -1.843608  5.180157
SkinThickness                     15.952218  0.109372 -0.520072
Insulin                          115.244002  2.272251  7.214260
BMI                                7.884160 -0.428982  3.290443
DiabetesPedigreeFunction           0.331329  1.919911  5.594954
Age                               11.760232  1.129597  0.643159
Outcome                            0.476951  0.635017 -1.600930
univariate analysis of sklearn diabetes dataset
                Mean      Median       Mode     Variance  Standard Deviation  \
age    -1.444295e-18    0.005383   0.016281     0.002268            0.047619
sex     2.543215e-18   -0.044642  -0.044642     0.002268            0.047619
bmi    -2.255925e-16   -0.007284  -0.030996     0.002268            0.047619
bp     -4.854086e-17   -0.005670  -0.040099     0.002268            0.047619
s1     -1.428596e-17   -0.004321  -0.037344     0.002268            0.047619
s2      3.898811e-17   -0.003819  -0.001001     0.002268            0.047619
s3     -6.028360e-18   -0.006584  -0.013948     0.002268            0.047619
s4     -1.788100e-17   -0.002592  -0.039493     0.002268            0.047619
s5      9.243486e-17   -0.001947  -0.018114     0.002268            0.047619
s6      1.351770e-17   -0.001078   0.003064     0.002268            0.047619
target  1.521335e+02  140.500000  72.000000  5943.331348           77.093005

        Skewness  Kurtosis
age    -0.231382 -0.671224
sex     0.127385 -1.992811
bmi     0.598148  0.095094
bp      0.290658 -0.532797
s1      0.378108  0.232948
s2      0.436592  0.601381
s3      0.799255  0.981507
s4      0.735374  0.444402
s5      0.291754 -0.134367
s6      0.207917  0.236917
target  0.440563 -0.883057
```

```
import matplotlib.pyplot as plt
def scatter_skew_kurtosis(df, title):
    skewness = df.skew(numeric_only=True)
    kurtosis = df.kurtosis(numeric_only=True)
    features = skewness.index
    plt.figure(figsize=(14,6))
    plt.scatter(features, skewness, marker='o', label='Skewness')
    plt.scatter(features, kurtosis, marker='x', label='Kurtosis')
    plt.xticks(rotation=60)
    plt.axhline(0)
    plt.title(f"Skewness vs Kurtosis - {title}")
    plt.xlabel("Features")
    plt.ylabel("Value")
    plt.legend()
    plt.tight_layout()
    plt.show()
```

```
scatter_skew_kurtosis(pima, "Pima Dataset")
```




```
scatter_skew_kurtosis(sklearn_df, "Sklearn Diabetes Dataset")
```



Frequency Distribution

```
print("Frequency of Outcome in Pima Dataset")
print(pima["Outcome"].value_counts())
```

```
Frequency of Outcome in Pima Dataset
Outcome
0    500
1    268
Name: count, dtype: int64
```

Bivariate analysis:Linear Regression

```
X = sklearn_df[['bmi']]
y = sklearn_df['target']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)
lin_reg = LinearRegression()
lin_reg.fit(X_train, y_train)
y_pred = lin_reg.predict(X_test)
sklearn_linear_r2 = r2_score(y_test, y_pred)
sklearn_linear_mse = mean_squared_error(y_test, y_pred)
print("Linear Regression Results (Sklearn Dataset)")
```

```
print("Coefficient:", lin_reg.coef_)
print("Intercept:", lin_reg.intercept_)
print("R2 Score:", sklearn_linear_r2)
print("MSE:", sklearn_linear_mse)
```

```
Linear Regression Results (Sklearn Dataset)
Coefficient: [988.41931249]
Intercept: 151.04202449316014
R2 Score: 0.2803417492440603
MSE: 3884.936720961032
```

Logistic Regression

```
X = pima[['Glucose']]
y = pima['Outcome']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)
log_reg = LogisticRegression()
log_reg.fit(X_train, y_train)
y_pred = log_reg.predict(X_test)
pima_logistic_acc = accuracy_score(y_test, y_pred)
print("Logistic Regression Results (Pima Dataset)")
print("Accuracy:", pima_logistic_acc)
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))
```

```
Logistic Regression Results (Pima Dataset)
Accuracy: 0.7229437229437229
Confusion Matrix:
 [[128  23]
 [ 41  39]]
```

Multiple Regression Analysis

```
X = sklearn_df.drop(columns=['target'])
y = sklearn_df['target']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)
multi_lin = LinearRegression()
multi_lin.fit(X_train, y_train)
y_pred = multi_lin.predict(X_test)
sklearn_multi_r2 = r2_score(y_test, y_pred)
print("Multiple Linear Regression (Sklearn Dataset)")
print("R2 Score:", sklearn_multi_r2)
print("MSE:", mean_squared_error(y_test, y_pred))
```

```
Multiple Linear Regression (Sklearn Dataset)
R2 Score: 0.4772897164322617
MSE: 2821.750981001311
```

Multiple Logistic Regression

```
X = pima.drop(columns=['Outcome'])
y = pima['Outcome']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)
multi_log = LogisticRegression(max_iter=2000)
multi_log.fit(X_train, y_train)
y_pred = multi_log.predict(X_test)
print("Multiple Logistic Regression (Pima Dataset)")
print("Accuracy:", accuracy_score(y_test, y_pred))
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))
```

```
Multiple Logistic Regression (Pima Dataset)
Accuracy: 0.7359307359307359
Confusion Matrix:
 [[120  31]
 [ 30  50]]
```

MULTIPLE REGRESSION COMPARISON

```
def multiple_regression_pima(data):
    X = data[['BMI','Age','Insulin','SkinThickness']]
    y = data['Glucose']
    X_train,X_test,y_train,y_test = train_test_split(X,y,test_size=0.3,random_state=42)
    model = LinearRegression()
    model.fit(X_train,y_train)
    pred = model.predict(X_test)
    mse = mean_squared_error(y_test,pred)
    return mse
def multiple_regression_sklearn(data):
    X = data.drop(columns=['target'])
    y = data['target']
```

```
    X_train,X_test,y_train,y_test = train_test_split(X,y,test_size=0.3,random_state=42)
    model = LinearRegression()
    model.fit(X_train,y_train)
    pred = model.predict(X_test)
    mse = mean_squared_error(y_test,pred)
    return mse
pima_multi_mse = multiple_regression_pima(pima)
sklearn_multi_mse = multiple_regression_sklearn(sklearn_df)
print("Pima Dataset Multiple Regression MSE (predicting Glucose):", pima_multi_mse)
print("Sklearn Diabetes Dataset Multiple Regression MSE (predicting target):", sklearn_multi_mse)
```

```
Pima Dataset Multiple Regression MSE (predicting Glucose): 795.3429239231998
Sklearn Diabetes Dataset Multiple Regression MSE (predicting target): 2821.750981001311
```

COMPARISON TABLE

```
comparison = pd.DataFrame({
    "Metric":["Logistic Accuracy","Linear R2","Linear MSE","Multiple Regression MSE"],
    "Sklearn Diabetes Dataset":["N/A", sklearn_linear_r2, sklearn_linear_mse, sklearn_multi_mse],
    "Pima Dataset":[pima_logistic_acc, "N/A", "N/A", pima_multi_mse]
})
comparison
```

	Metric	Sklearn Diabetes Dataset	Pima Dataset
0	Logistic Accuracy	N/A	0.722944
1	Linear R2	0.280342	N/A
2	Linear MSE	3884.936721	N/A
3	Multiple Regression MSE	2821.750981	795.342924